Solving Simple Equations Using Bar Models

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Bar Model

A **bar model** is a visual representation that uses rectangular bars to show quantities and the relationships between them.



Unknown

An **unknown** is a quantity whose value is not yet known.
We often represent it using a variable such as x.



The bar says: "I do not know this value yet."

Equation

An **equation** is a mathematical statement showing that two expressions are equal.

$$x + 7 = 18$$

The left side and the right side have the same value.

Equal Parts

Equal parts represent quantities with the same value.

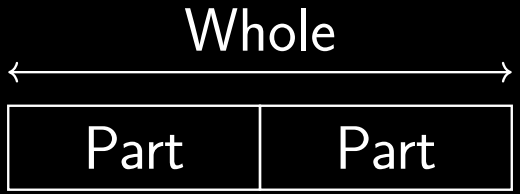


Every bar is the same size, so every bar has the same value.

Whole and Part

A **whole** represents the total quantity.

A **part** represents a portion of that total.



Memory Aid

Part + Part = Whole

Whole − Part = Other Part

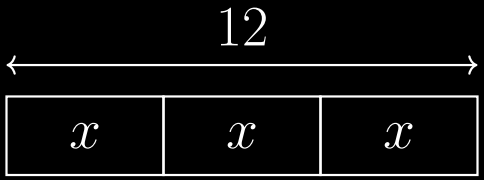
Finding Each Equal Part

When the whole and the number of equal parts are known, **divide** the whole by the number of parts to find the value of each part.

$$\text{each part} = \text{whole} \div \text{number of parts}$$

Finding Each Equal Part: Bar Model

Whole $= 12$, with 3 equal parts.



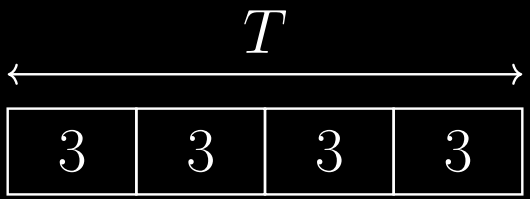
Finding the Whole

When the value of each equal part is known, **multiply** it by the number of parts to find the whole.

$$\text{whole} = \text{value of each part} \times \text{number of parts}$$

Finding the Whole: Bar Model

4 equal parts, each worth 3.



$$T = 4 \times 3 = 12$$

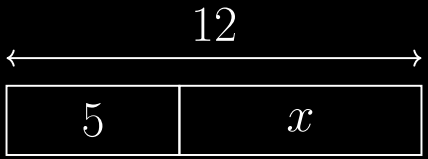
Finding a Missing Part

When one part is unknown, **subtract** the known part from the whole to find the missing part.

$$\text{missing part} = \text{whole} - \text{known part}$$

Finding a Missing Part: Bar Model

Whole $= 12$. One part is 5. The other part is x.



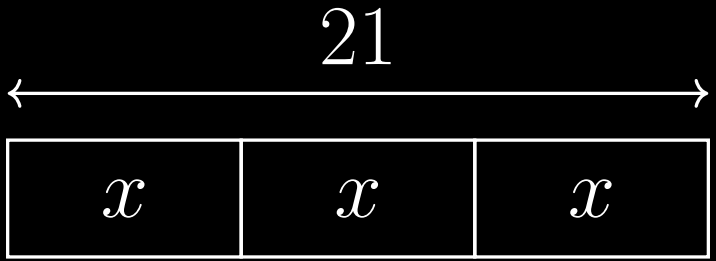
Which Operation Do I Use?

- **I know the whole and the number of parts:** I find each part by **dividing**.
- **I know each part and the number of parts:** I find the whole by **multiplying**.
- **I know the whole and one part:** I find the other part by **subtracting**.

Examples

Example 1

A bar model has three equal parts with a total value of 21. Find x.



Example 1: Solution

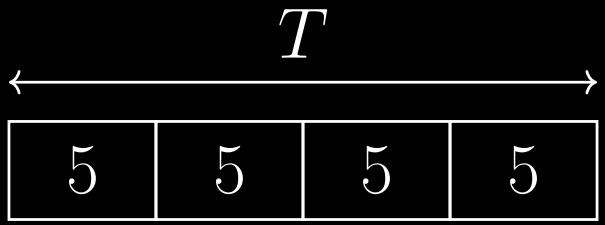
$$3x = 21$$

$$x = 21 \div 3 = 7$$

Check: $3 \times 7 = 21$

Example 2

A bar model shows four equal parts, each with a value of 5. Find the total value T.



Example 2: Solution

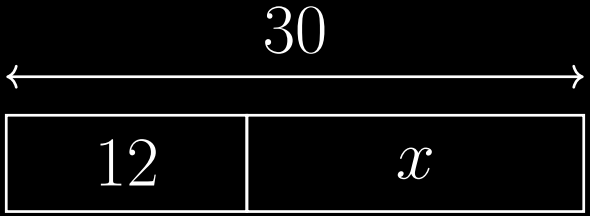
$$T = 4 \times 5$$

$$T = 20$$

Check: $5 + 5 + 5 + 5 = 20$

Example 3

A bar model represents a total of 30 divided into two parts. One part is 12, and the other part is x. Find x.



Example 3: Solution

$$12 + x = 30$$

$$x = 30 - 12 = 18$$

Check: $12 + 18 = 30$

Practice Exercises

Practice 1

Write an equation that represents the model, then solve for the unknown.

A bar model represents a total of 18 divided into two parts. One part is 7, and the other part is x.

Practice 2

Write an equation that represents the model, then solve for the unknown.

A bar model has three equal parts with a total value of 27. Find the value of x.

Practice 3

Write an equation that represents the model, then solve for the unknown.

A bar model shows four equal parts, each with a value of 6. Find the total value T.

Practice 4

Write an equation that represents the model, then solve for the unknown.

A bar model has five equal parts with a total value of 45. Find the value of each part, x.

Practice 5

Write an equation that represents the model, then solve for the unknown.

A bar model shows two equal parts with a total value of 50. Find the value of each part, x.

Practice 6

Write an equation that represents the model, then solve for the unknown.

A bar model has three equal parts, each worth 12. Find the total value T.

Practice 7

Write an equation that represents the model, then solve for the unknown.

A bar model shows six equal parts with a total value of 54. Find the value of each part, x.

Activities

Activity 1

Write an equation that represents the model, then solve for the unknown.

A bar model has four equal parts with a total value of 36. Find the value of each part, x.

Activity 2

Write an equation that represents the model, then solve for the unknown.

A bar model shows five equal parts, each with a value of 7. Find the total value T.

Activity 3

Write an equation that represents the model, then solve for the unknown.

A bar model represents a total of 48. It has two parts: one is x, and the other is 19. Find x.

Activity 4

Write an equation that represents the model, then solve for the unknown.

A bar model has three equal parts with a total value of 33. Find the value of each part, x.

Activity 5

Write an equation that represents the model, then solve for the unknown.

A bar model shows two equal parts with a total value of 64. Find the value of each part, x.

Activity 6

Write an equation that represents the model, then solve for the unknown.

A bar model has four equal parts, each worth 9. Find the total value T.

Activity 7

Write an equation that represents the model, then solve for the unknown.

A bar model shows seven equal parts with a total value of 63. Find the value of each part, x.

Thank you!